Learning Analytics: Formula Cheat Sheet

NPTEL exam revision

1. Classification metrics (TP, TN, FP, FN)

- **Accuracy** = (TP+TN) / (TP+TN+FP+FN)
- **Precision (PPV)** = TP / (TP+FP)
- **Recall = Sensitivity = TPR** = TP / (TP+FN)
- **Specificity (TNR)** = TN / (TN+FP)
- **FPR** = FP / (FP+TN) = 1 − Specificity
- **FNR** = FN / (FN+TP) = 1 − Recall
- **NPV** = TN / (TN+FN)
- **F1** = 2·P·R / (P+R) = 2TP / (2TP+FP+FN)
- **F_β** = $(1+\beta^2)\cdot P\cdot R / (\beta^2\cdot P + R)$
- **Balanced accuracy** = (Sensitivity + Specificity) / 2
- **Cohen's Kappa**: $\kappa = (p_o - p_e) / (1 - p_e)$
  p_o = observed agreement, p_e = agreement expected by chance
- **AUC-ROC**: area under TPR vs FPR curve (0.5 = random, 1 = perfect)

2. Correlation

- **Pearson**: $r = \Sigma(x_i-\bar{x})(y_i-\bar{y}) / \sqrt{[\Sigma(x_i-\bar{x})^2 \cdot \Sigma(y_i-\bar{y})^2]} = Cov(X,Y) / (\sigma_x\sigma_y)$
- **Spearman** (no ties): $\rho = 1 - 6\Sigma d_i^2 / [n(n^2-1)]$, d_i = rank difference
- **Kendall's tau**: τ = (concordant − discordant) / [n(n−1)/2]
- **Coefficient of determination** = r^2
- Range of r, ρ, τ: −1 to +1 (0 = no linear / monotonic relation)

3. Regression

- **Linear**: $y = \beta_0 + \beta_1 x + \varepsilon$
- $\beta_1 = \Sigma(x-\bar{x})(y-\bar{y}) / \Sigma(x-\bar{x})^2$; $\beta_0 = \bar{y} - \beta_1\bar{x}$
- **MSE** = $(1/n)\Sigma(y_i-\hat{y}_i)^2$; **RMSE** = √MSE
- **MAE** = $(1/n)\Sigma|y_i-\hat{y}_i|$
- **R^2** = $1 - SS_{res}/SS_{tot}$
- **Adjusted R^2** = $1 - (1-R^2)(n-1)/(n-k-1)$
- **Logistic**: $P(y=1) = 1 / (1 + e^{-(\beta 0+\beta 1 x)})$
  odds = p/(1−p); log-odds (logit) = $\beta_0 + \beta_1 x$

4. Descriptive statistics

- **Mean** = Σx / n
- **Variance** (sample) = $\Sigma(x-\bar{x})^2 / (n-1)$; **SD** = √variance
- **z-score** = (x − μ) / σ
- **Min-max normalization** = (x − min) / (max − min)
- **Covariance** = $\Sigma(x-\bar{x})(y-\bar{y}) / (n-1)$
- **IQR** = Q3 − Q1; outlier if < Q1 − 1.5·IQR or > Q3 + 1.5·IQR

5. Clustering / distance

- **Euclidean** = $\sqrt{\Sigma(x_i-y_i)^2}$
- **Manhattan** = $\Sigma|x_i-y_i|$
- **Cosine similarity** = $A\cdot B / (\|A\|\,\|B\|)$
- **Jaccard** = $|A\cap B| / |A\cup B|$
- **K-means objective (WCSS)** = $\Sigma_k \Sigma_{x\in k} \|x - \mu_k\|^2$
- **Silhouette** = (b − a) / max(a, b)
  a = mean intra-cluster distance, b = mean nearest-other-cluster distance

6. Probability, Bayes, trees

- **Bayes**: $P(A|B) = P(B|A)\cdot P(A) / P(B)$
- **Entropy** = $-\Sigma\, p_i \log_2 p_i$
- **Information gain** = Entropy(parent) − $\Sigma\, (n_i/n)\cdot$Entropy($child_i$)
- **Gini** = $1 - \Sigma\, p_i^2$

7. Association rules

- **Support**(A→B) = P(A∩B)
- **Confidence** = P(A∩B) / P(A)
- **Lift** = Confidence / P(B) (> 1 = positive association)

8. Knowledge tracing & educational models

- **BKT update**: $P(L_{n+1}) = P(L_n|obs) + (1 - P(L_n|obs))\cdot P(T)$
  $P(L_0)$ = prior, P(T) = learn, P(G) = guess, P(S) = slip
- **BKT**: $P(correct) = P(L)(1 - P(S)) + (1 - P(L))\cdot P(G)$
- **IRT 1PL (Rasch)**: $P(correct) = 1 / (1 + e^{-(\theta-b)})$
- **2PL**: $P = 1 / (1 + e^{-a(\theta-b)})$
- **3PL**: $P = c + (1-c) / (1 + e^{-a(\theta-b)})$
  θ = ability, b = difficulty, a = discrimination, c = guessing
- **Learning curve (power law)**: $Y = a\cdot X^b$

9. Text / network analytics, reliability

- **TF** = count(t,d) / words(d); **IDF** = log(N / df); **TF-IDF** = TF·IDF
- **Degree centrality** = degree / (n−1)
- **Density** = 2E / [n(n−1)] (undirected)
- **Cronbach's α** = $[k/(k-1)]\cdot(1 - \Sigma\sigma_i^2 / \sigma_t^2)$

10. Hypothesis testing

- **t-statistic** = $(\bar{x} - \mu) / (s/\sqrt{n})$
- **Chi-square**: $\chi^2 = \Sigma\ (O - E)^2 / E$
- **Cohen's d** = $(M_1 - M_2)$ / pooled SD

Quick tips

- Imbalanced data → prefer F1 / precision / recall over accuracy.
- Precision matters when FP is costly; recall when FN is costly.
- Use Spearman for ordinal or non-linear but monotonic data; Pearson for linear.